

INTRODUCTION

This report provides a comprehensive analysis of the AdventureWorksLT2025 sales database to evaluate revenue performance, product profitability, and customer value. The analysis addresses five key areas: monthly revenue trends, product revenue concentration, repeat versus one-time buyer behaviour, customer lifetime value segmentation into quartiles, and category-level revenue, margin, and unit sales. Using SQL queries on SalesLT tables, the study aims to identify high-value products, customers, and categories, assess retention and concentration risks, and provide data-driven recommendations for pricing, retention, and growth.

1.Customer LTV Segmentation

Analysis of 32 customers segmented into 4 quartiles by lifetime value shows revenue concentration in the top quartile. Q1 contains 8 customers with LTVs from \$63,686.27 to \$119,960.82, led by Terry Eminhizer at \$119,960.82, followed by Krishna Sunkammurali and Christopher Beck. All customers have a single order, so LTV equals first-order value, with a significant drop between Q1 and Q2 indicating a distinct high-value tier and dependence on a small number of large one-time purchases.

2.Product Category Profitability

Revenue is concentrated in three bike categories: Touring Bikes \$220,655 [252 units], Road Bikes \$183,130 [222 units] and Mountain Bikes \$170,825 [209 units], which together drive over 80% of total revenue. However, gross margin analysis shows negative profitability for top revenue drivers, with Road Bikes at -10.17% [-\$18,616] and Touring Bikes at -4.15% [-\$9,147]. Positive margin is driven by low-revenue accessory categories, including Shorts 36.54%, Helmets 35.71% and Vests 33.32%, while unit sales are highest for Bikes and Jerseys [230 units] despite a 29x difference in per-unit revenue value.

3.Monthly Sales Trend

Revenue analysis by month using DATETRUNC(MONTH) shows a single reporting period in the current dataset: June 2008 with total revenue of \$956,303.59. Month-over-month growth is not calculable as prior period revenue is NULL due to single-month data availability. The dataset therefore represents a point-in-time snapshot; a multi-month history is required to identify periods of strongest growth or decline.

4.Top Products & Cumulative Revenue

Analysis of 142 selling products shows revenue is moderately concentrated in premium bike models. The top 1 product, Touring-1000 Blue, 60, generates \$37,191.49, representing 5.25% of total revenue. The top 5 products account for 24.0% and the top 11 products account for 40.97% of total revenue. Approximately 25-30% of SKUs drive 80% of revenue, with Touring, Mountain-200 and Road-350 models dominating, indicating a Pareto effect where a small number of high-value bikes account for the majority of revenue.

5.Customer Retention

Retention analysis of 32 customers shows 100% are One-Time Buyers, generating 100% of total revenue (\$956,303.59). No Repeat Buyers were identified, contributing 0% of customers and \$0 revenue. This indicates a purely transactional customer base with no repeat purchase behaviour in the observed period, highlighting a critical retention risk and the need for strategies to drive second purchases.

RECOMMENDATIONS

- **Fix Data Scope:** Restore full historical data for AdventureWorksLT2025 - current analysis is only 1 month [June 2008, 32 orders] so MoM trend cannot be calculated. Expand dataset to enable real trend and cohort analysis.
- **Fix Pricing & Margin:** Audit Product.StandardCost - Road Bikes [-10.17%, - \$18,616] and Touring Bikes [-4.15%, -\$9,147] are selling below cost. Set floor price at Standard Cost + 15% and reduce discounting on bikes.
- **Protect Top Products:** Top 11 products drive 40.97% of revenue. Ensure 100% availability, prioritize marketing, and protect stock for Touring-1000 Blue 60 [\$37,191, 5.25%], Touring-1000 Yellow 60, and Mountain-200 series.
- **Rationalize Long Tail:** ~110 products contribute <60% revenue. Review bottom 50% of 142 products for discontinuation to reduce carrying costs.
- **Address 100% One-Time Buyer Risk:** 32 customers = 100% one-time, \$0 repeat revenue. Launch post-purchase nurture - service emails, accessory offers, and loyalty program to drive second purchase.

- **VIP Customer Retention:** Protect Q1 8 customers [\$63,686 - \$119,960, led by Terry Eminhizer \$119,960] who drive >65% revenue. Implement account management, exclusive offers, and referral incentives to prevent churn.
- **Drive High-Margin Mix:** Push accessories with 26-36% margin at bike checkout - Shorts [36.54%], Helmets [35.71%], Vests [33.32%], Crank sets and Pedals [26.00%] to offset negative bike margins.
- **Increase AOV:** Bundle high-unit, low-value items like Jerseys [230 units, \$30/unit] with bikes to convert unit volume into revenue and improve margin mix per transaction.

Conclusion

In conclusion, analysis of the AdventureWorksLT2025 snapshot of 32 orders and 32 customers totalling \$956,303 in June 2008 reveals a high-value, single-purchase business with significant concentration risk. Revenue is driven by a small number of premium bike models and three bike categories accounting for over 80% of sales, with the top 11 products contributing 40.97% and the top 8 customers contributing the majority of revenue, led by Terry Eminhizer at \$119,960. However, the business shows zero repeat buyers and negative gross margins on top revenue drivers including Road Bikes at -10.17% and Touring Bikes at -4.15%, while true profitability is sustained by low-volume accessories such as Shorts at 36.54% and Helmets at 35.71%. The findings highlight the need to address below-cost pricing, improve customer retention, and leverage high-margin attachments to ensure sustainable growth.